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Stochastic Processes

Lecture 24

Karthik P. N.
Assistant Professor, Department of AI
Email: pnkarthik@ai.iith.ac.in

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Martingale

Fix a probability space $(\Omega, \mathscr{F}_0, \mathbb{P})$.

Definition (Martingale)

Let $\mathscr{G} = \{\mathscr{G}_n\}_{n\in\mathbb{N}}$ be any filtration, and let $\mathbf{X} = \{X_n\}_{n\in\mathbb{N}}$ be a process such that

1. $\mathbb{E}[|X_n|] < +\infty$ for all n.
2. X_n is $\mathscr{G}_n$-measurable for each n.

Then, $\mathbf{X}$ is called a **martingale with respect to the filtration $\mathscr{G}$** if

$$\mathbb{E}[X_{n+1} \mid \mathscr{G}_n] \;=\; X_n \qquad \forall\, n \in \mathbb{N}.$$

- If the condition
$$\mathbb{E}[X_{n+1} \mid \mathscr{G}_n] \;\geq\; X_n \qquad \forall\, n \in \mathbb{N}$$
holds, then $\mathbf{X}$ is called a **submartingale**
- If the condition
$$\mathbb{E}[X_{n+1} \mid \mathscr{G}_n] \;\leq\; X_n \qquad \forall\, n \in \mathbb{N}$$
holds, then $\mathbf{X}$ is called a **supermartingale**

Remarks

- A process $\mathbf{X}$ is a supermartingale if and only if $-\mathbf{X}$ is a submartingale
- A process $\mathbf{X}$ is a martingale if and only if it is both a supermartingale and a submartingale
- If $\mathbf{X} = \{X_n\}_{n\in\mathbb{N}}$ is a martingale adapted to $\{\mathscr{G}_n\}_{n\in\mathbb{N}}$, then

$$\mathbb{E}[X_1 - X_1 \mid \mathscr{G}_1] \;=\; 0, \qquad \mathbb{E}[X_n - X_1 \mid \mathscr{G}_n] \;=\; X_{n-1} - X_1 \qquad \forall\, n \geq 2,$$

  thus proving that $\{X_n - X_1\}_{n\in\mathbb{N}}$ is also a martingale that is null at $n = 1$
- If $\mathbf{X} = \{X_n\}_{n\in\mathbb{N}}$ is a martingale adapted to $\{\mathscr{G}_n\}_{n\in\mathbb{N}}$, then for any $m < n$,

$$\mathbb{E}[X_n \mid \mathscr{G}_m] \;=\; \mathbb{E}\Big[\mathbb{E}[X_n \mid \mathscr{G}_{n-1}] \mid \mathscr{G}_m\Big] \;=\; \mathbb{E}[X_{n-1} \mid \mathscr{G}_m] \;=\; \cdots \;=\; X_m.$$

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Examples

- **Sums of IID zero-mean RVs:**
  - Suppose $\{X_n\}_{n\in\mathbb{N}}$ are IID RVs with $\mathbb{E}[X_1] = 0$
  - Setting $S_0 = 0$, $S_n = \sum_{k=1}^{n} X_k$, $\mathscr{G}_0 = \{\emptyset, \Omega\}$, and $\mathscr{G}_n = \sigma(X_1, \ldots, X_n)$ for all $n \geq 1$, we get
  $$\mathbb{E}[S_n \mid \mathscr{G}_{n-1}] \; = \; S_{n-1} \qquad \forall\, n \in \mathbb{N}.$$
  Thus, $\{S_n\}_{n\geq 0}$ is a martingale with respect to $\{\mathscr{G}_n\}_{n\geq 0}$
- **Product martingale:**
  - Suppose $\{X_n\}_{n\in\mathbb{N}}$ are IID RVs with $\mathbb{E}[X_1] = 1$
  - Setting $M_0 = 1$, $M_n = \prod_{k=1}^{n} X_k$, $\mathscr{G}_0 = \{\emptyset, \Omega\}$, and $\mathscr{G}_n = \sigma(X_1, \ldots, X_n)$ for all $n \geq 1$, we get
  $$\mathbb{E}[M_n \mid \mathscr{G}_{n-1}] \; = \; M_{n-1} \qquad \forall\, n \in \mathbb{N}.$$
  Thus, $\{M_n\}_{n\geq 0}$ is a martingale with respect to $\{\mathscr{G}_n\}_{n\geq 0}$
- **Accumulating data about a RV:**
  - Given a filtration $\{\mathscr{G}_n\}_{n\in\mathbb{N}}$ and a RV X with $\mathbb{E}[|X|] < +\infty$, let $M_n := \mathbb{E}[X \mid \mathscr{G}_n]$ for all n
  - Using tower property, it can be shown that $\{M_n\}_{n\in\mathbb{N}}$ is a martingale with respect to $\{\mathscr{G}_n\}_{n\in\mathbb{N}}$

Predictable Sequence and Gambling

- Imagine that you are gambling sequentially in rounds
- Suppose that you bet 1 unit of money each round, and your earnings after n rounds is $X_n \quad (X_0 = 0)$
- $X_n - X_{n-1}$ is your earnings in round n per unit of money bet
- For each k, if H_k amount of money is bet in round k, then the total earnings after n rounds is given by

$$(\mathbf{H} \cdot \mathbf{X})_n \;:=\; \sum_{k=1}^{n} H_k \left(X_k - X_{k-1}\right).$$

- H_n is, in general, a function of $(H_1, \ldots, H_{n-1})$ and $(X_1, \ldots, X_{n-1})$
  The process $\mathbf{H} = \{H_n\}_{n\in\mathbb{N}}$ is called a **predictable sequence**

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Predictable Sequence and Gambling

- **The martingale gambling system:**
  - Suppose that the process $\mathbf{X} = \{X_n\}_{n \geq 0}$ satisfies

    $$X_0 = 0, \qquad \underbrace{X_1 - X_0}_{\xi_1}, \ \underbrace{X_2 - X_1}_{\xi_2}, \ \ldots \ \text{IID}, \qquad \mathbb{P}(\{\xi_1 = 1\}) = p = 1 - \mathbb{P}(\{\xi_1 = -1\}).$$

  - The martingale gambling system is defined by the predictable sequence $\mathbf{H} = \{H_n\}_{n \in \mathbb{N}}$ given by

    $$H_n = \begin{cases} 2H_{n-1}, & \xi_{n-1} = -1, \\ 1, & \xi_{n-1} = +1. \end{cases}$$

  - **Strategy: double the bet when you lose**
  - Although counterintuitive, the above strategy is quite promising: if we lose k times and then win, our net earnings will be 1

H_n	1	2	4	8	16
ξ_n	-1	-1	-1	-1	1
$(H \cdot X)_n$	-1	-3	-7	-15	1

You Cannot Beat the System!

Theorem (You Cannot Beat the System)

Let $\mathscr{G}_0 = \{\emptyset, \Omega\}$ and let $\mathscr{G} = \{\mathscr{G}_n\}_{n \in \mathbb{N}}$ be a filtration. Assume that $X_0 = 0$.
If $\mathbf{X} = \{X_n\}_{n \geq 0}$ is a **supermartingale**, *and $\mathbf{H} = \{H_n\}_{n \in \mathbb{N}}$ is* **non-negative and bounded predictive process** *w.r.t. $\mathscr{G}$ (i.e., $|H_n| \leq L$ for all n, for some $L < +\infty$, and H_n is $\mathscr{G}_{n-1}$-measurable for all n), then*

$$\{(\mathbf{H} \cdot \mathbf{X})_n\}_{n \geq 0} \text{ is a supermartingale,}$$

with the convention that $(\mathbf{H} \cdot \mathbf{X})_0 = 0$.

Proof of Theorem 1:

- Note that for any n,

$$\mathbb{E}[(\mathbf{H} \cdot \mathbf{X})_{n+1} \mid \mathscr{G}_n] = (\mathbf{H} \cdot \mathbf{X})_n + \mathbb{E}[H_{n+1}(X_{n+1} - X_n) \mid \mathscr{G}_n] = (\mathbf{H} \cdot \mathbf{X})_n + H_{n+1}\, \mathbb{E}[X_{n+1} - X_n \mid \mathscr{G}_n] \leq (\mathbf{H} \cdot \mathbf{X})_n.$$

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Gambling up to a Stopping Time

- Let $\mathscr{G} = \{\mathscr{G}_n\}_{n \geq 1}$ be a filtration, and let $\mathscr{G}_0 = \{\emptyset, \Omega\}$
- Let τ be a stopping time adapted to $\mathscr{G}$ (i.e., τ takes values ≥ 1)
- Suppose that you bet 1 unit of money till τ and quit playing after τ rounds
- Then, the gambling strategy $\mathbf{H} = \{H_n\}_{n \in \mathbb{N}}$ is given by

$$H_n = \mathbf{1}_{\{n \leq \tau\}}, \qquad n \in \mathbb{N}.$$

  Notice that H_n is $\mathscr{G}_{n-1}$-measurable for all n

- In this case, we have

$$(\mathbf{H} \cdot \mathbf{X})_n = X_{\tau \wedge n} - X_0, \qquad \tau \wedge n := \min\{\tau, n\}.$$

  Notice that $(\mathbf{H} \cdot \mathbf{X})_0 = 0$

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Stopped Supermartingale

Theorem (Stopped Supermartingale)

Let $\mathscr{G}_0 = \{\emptyset, \Omega\}$, and let $\mathscr{G} = \{\mathscr{G}_n\}_{n\geq 1}$ be a filtration.
Let $\mathbf{X} = \{X_n\}_{n\geq 0}$ be a supermartingale with respect to $\mathscr{G} = \{\mathscr{G}_n\}_{n\geq 0}$.
Let τ be a stopping time adapted to $\mathscr{G}$.
Then, $\{X_{\tau\wedge n}\}_{n\geq 0}$ is a supermartingale with respect to $\{\mathscr{G}_n\}_{n\geq 0}$.

Note: *The process $\{X_{\tau\wedge n}\}_{n\geq 0}$ is called the* **stopped supermartingale**.

Proof of Theorem 2:

- If $\mathbf{X} = \{X_n\}_{n\geq 0}$ is a supermartingale w.r.t. $\{\mathscr{G}_n\}_{n\geq 0}$, then by Theorem 1, it follows that $\{X_{\tau\wedge n} - X_0\}_{n\geq 0}$ is a supermartingale w.r.t. $\{\mathscr{G}_n\}_{n\geq 0}$
- Because X_0 is $\mathscr{G}_0$-measurable, it is a constant RV
  Hence, the constant sequence $Y_n = X_0$ for all n trivially a supermartingale
- Because sum of two supermartingales is a supermartingale, the result follows